

Project Design Phase
Solution Architecture

Date	16 June 2025
Team ID	LTVIP2025TMID47387
Project Name	Toycraft tales: tableau's vision into toy manufacturer data
Maximum Marks	4 Marks

Solution Architecture:

Goals of the Architecture

1. ☒ **Find the best tech solution** to visualize how product position, price, promotions, and customer demographics affect sales volume.
2. ☒ **Define the system's structure and behavior** using Tableau Public for dashboards and Python/Tableau for preprocessing.
3. ☒ **Outline features, components, and development phases** clearly for visualization and interaction.
4. ☒ **Provide technical specifications** for implementation and scaling.

Architecture Components

Layer Component Description

Data LayerSource: Product Positioning.csv file with fields like Sales Volume, Foot Traffic,
Demographics, Promotion, etc.

Processing Layer Data cleaning, formatting, and calculated fields in Tableau or via Python (optional).

Application

LayerTableau logic engine for filters, calculations, and chart rendering. interactive
visuals: bar, funnel, donut, **Presentation Layer** waterfall, etc.
Final dashboard built in Tableau with

UsersRetail managers, marketing executives, analysts – accessing via Tableau Public or PDF
reports.

Development Phases

Phase Description

Phase 1 – Data Prep Clean CSV data, convert data types, handle missing values **Phase 2** – Basic
Visuals Create bar, donut, and filterable dashboards **Phase 3** – Advanced Visuals Funnel, area,
waterfall, and highlight tables with calculated fields **Phase 4** – Dashboard + Story Combine
visuals, enable filters, and build an interactive story **Phase 5** – Final Report Export dashboard
and insights for stakeholders